

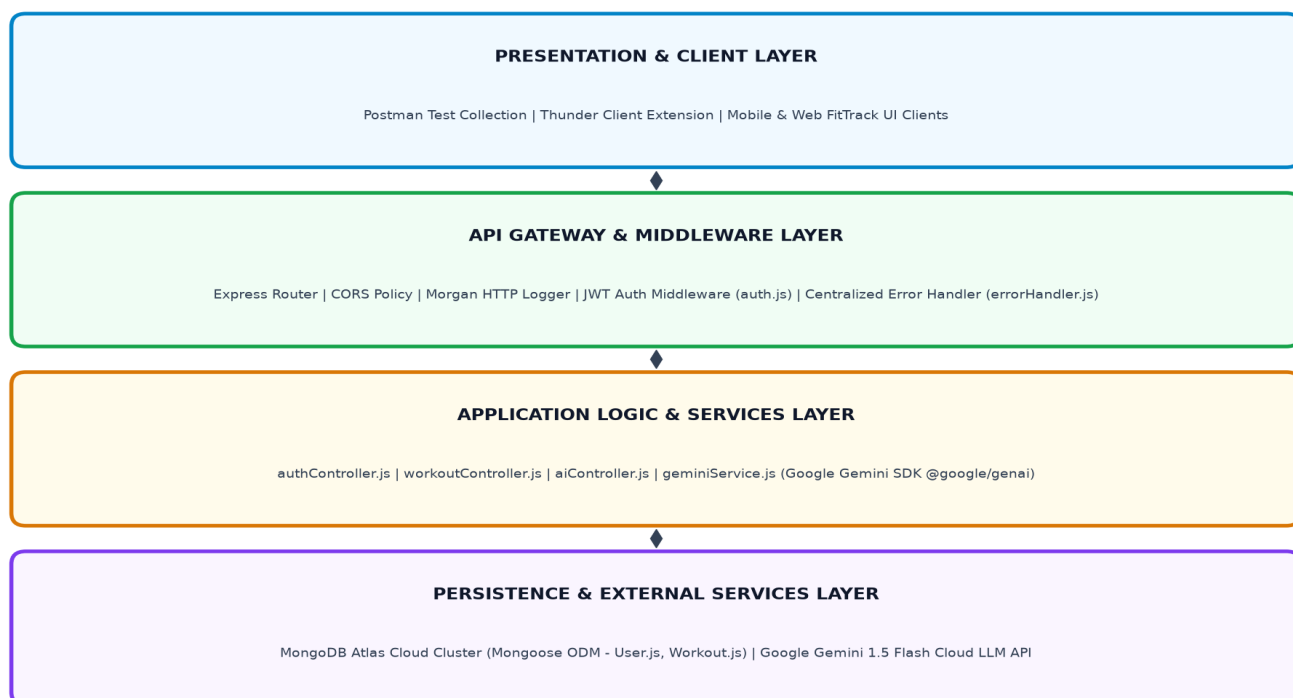
Solution Architecture Diagram

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	5 Marks

Solution Architecture Blueprint:

AI FitTrack API is engineered using a decoupled, 4-tier architecture separating presentation, gateway routing, application logic, and cloud persistence services.

- **Stateless REST Design:** Every request is self-contained with JWT Bearer authentication.
- **Decoupled AI Engine:** geminiService isolates LLM communication from HTTP route handlers.
- **Centralized Resilience:** Single error middleware catches synchronous and asynchronous failures.
- **Strict User Scoping:** All workout queries are strictly scoped to authenticated user context.



Solution Architecture — Component Description Table

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Architectural Layer Component Specifications:

Component Name	Layer	Role & Responsibilities	Technologies Used
Presentation Client	Presentation	Sends REST HTTP JSON requests (Auth, Workout CRUD, Search, AI). Manages test collections and bearer authorization tokens.	Postman v2.1, Thunder Client, Mobile/Web Client UI
API Gateway & Server	Gateway	Initializes Express application, mounts CORS policy, JSON body parser, Morgan HTTP logging, and environment variable configuration.	Node.js, Express.js (app.js, server.js), cors, morgan, dotenv
Security & Middleware	Middleware	Enforces Bearer token extraction and verification; rejects unauthenticated requests with 401; handles centralized errors.	jsonwebtoken, bcryptjs, middleware/auth.js, middleware/errorHandler.js
Controllers & Services	Application Logic	Implements business logic for auth, workout operations, regex search, and Gemini AI prompt formatting.	authController.js, workoutController.js, aiController.js, geminiService.js
External AI Service	External AI	Executes generative LLM queries to produce tailored workout plans and progress insights.	Google Gemini API, gemini-1.5-flash, @google/genai SDK
Data Storage Layer	Persistence	Stores and indexes Users and Workouts collections. Enforces schema constraints, default timestamps, and user ObjectId references.	MongoDB Atlas Cloud Cluster, Mongoose ODM (User.js, Workout.js, db.js)

Security Boundaries & Fault Tolerance:

- **Credential Isolation:** MongoDB connection strings and Google Gemini API keys reside exclusively in .env and are never committed to git.
- **Resource Isolation:** A user cannot view, modify, or delete workouts logged by other users (returns 404/403).
- **Graceful Fallback:** If GEMINI_MODEL is omitted, geminiService automatically falls back to gemini-1.5-flash for dependable uptime.